MILITARY SPECIFICATION

SEPARATORS, OIL-WATER, COALESCER TYPE: FILTER ELEMENTS, FLUID PRESSURE, COALESCER, OILY WASTEWATER, EXPENDABLE: AND PREFILTER ELEMENTS, OILY WASTEWATER, EXPENDABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers oil-water separators, filter-coalescer elements, and prefilter elements used in fixed or portable, land-based or shipboard oil pollution control systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-S-766	- Steel Plates, Sheets and Strip - Corrosion Resisting.
VV-F-800	- Fuel Oil, Diesel.
VV-G-76	- Gasoline, Automotive.
WW-P-406	- Pipe, Steel (Seamless and Welded) (For Ordinary Use).
WW-P-521	- Pipe Fittings, Flange Fittings, and Flanges Steel and Malleable Iron (Threaded and Butt-Welding) 150 Pound.
WW-T-799	- Tube, Copper, Seamless, Water and Refrigeration (For use with Solder-Flared or Compression Type Fittings).
WW-V-51	- Valve, Angle, Check, and Globe, Bronze (125, 150 and 200 Pound, Threaded End, Flange Ends, Solder Ends, and Bronzed End, For Land Use).

FSC 4330

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Military

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
JAN-P -700	- Paint, Deck, Gray, Inside.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-9000	- Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-A-18001	- Anodes, Corrosion Preventive, Zinc; Slab, Disc and Rod Shaped.
MIL-I-18997	- Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-P-23236	- Paint Coating Systems, Steel Ship Tank, Fuel and Salt Water Ballast.
MIL-F-24397	- Fuel, Navy Distillate.
MIL-F-52847	- Filter Elements, Fluid Pressure, Coalescer, Oily Wastewater, Expendable: And Prefilter Elements, Oily Wastewater, Expendable.

STANDARDS

Military

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
KtL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the 'contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Sections VIII and IX, Pressure Vessels, Division 1, and Welding Qualifications.

(Application for copies should be addressed to American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

Standard No. 45 - Recommended Practice for Electric Instillations on Shipboard.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION

Volume 5, Article 500 - Hazardous Locations.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Sreet, Boston, MA 02110.)

COUNTY OF LOS ANGELES -AIR POLLUTION CONTROL DISTRICT

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the County of Los Angeles - Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013)

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AMERICAN WELDING SOCIETY INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1331 - Surface and Interracial Tension of Solutions of Surface-Active Agents.
D2276 - Particulate Contaminant in Aviation Turbine Fuels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3• REQUIREMENTS

3.1 Description. The oil-water separators (hereinafter called "separators") shall consist of one or more filter-coalescer vessels connected in series and preceeded by a vertical prefilter utilizing expendable filter-coalescer elements (hereinafter called "elements") and expendable prefilter elements (hereinafter called "refilters"). The separators shall be provided with an automatic oil discharge system connected to one or more of the stages, an automatic water discharge control valve, automatic air eliminator on the prefilter vessel, pressure gages for indicating pressure drop across each vessel, oil-level sight gages on each vessel, water drain valves on the bottom of each vessel, and sampling valves on the inlet to each vessel and on the water and oil discharge lines (see 6.3).

3.1.1 Design. The detailed design configuration and flow rate of the separators shall be as specified in the individual detailed specifications or in the end item specification. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.2 First article (reproduction model). The supplier shall furnish a separator for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods will produce separators that comply with requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.4). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests (see 6.2).

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3.3 Materials. Materials shall be as specified herein or as specified in the individual detailed specification or end item specification. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Sheet steel. Sheet steel used in pressure vessel construction shall conform to QQ-S-766. Pressure vessels shall be epoxy coated inside and outside in accordance with MIL-P-23236, Type I, Class 1, except as specified in 3.3.5

3.3.2 Pipe and tubing. All piping shall conform to WW-P-406, Weight A, Class 1; copper tubing shall conform to WW-T-799, Type K, Temper Soft, Class 1.

3.3.3 Pipe fittings. Standard weight ferrous fittings (150-pound WPS) shall be malleable iron, and shall conform to WW-P-521, Type I.

3.3.4 Valves. Globe, angle, and check valves shall be bronze and shall conform to WW-V-51, Class A.

3.3.5 Interior components pressure vessels. All bolts, threaded rods, washers, nuts, and element and prefilter sealing surfaces in contact with water shall be 316 stainless steel and shall be left uncoated.

3.3.6 Rubber gaskets, O-rings, and diaphragms. All rubber gaskets, O-rings, and diaphragms in contact with oils or oily wastewater shall be of NBR rubber or equal.

3.3.7 Anodes, zinc. Sacrificial zinc anodes 1/2-inch in diameter and 2-inches in length conforming to MIL-A-18001, Class 2, Type ZRN shall be installed in the bottom of each pressure vessel under the element and prefilter support. The anodes shall be serviceable from outside of the pressure vessel.

3.3.8 Screw threads. All screw threads required shall be in accordance with Handbook H28.

3.4 Environmental conditions.

3.4.1 Operating temperatures. Unless otherwise specified (see 6.2), the separators, elements, and prefilters, when assembled, shall permit rated flow in any ambient temperature in the range of plus 125° F to plus 33° F.

3.4.2 Storage temperatures. Unless otherwise specified (see 6.2), the separators, elements, and prefilters shall not be damaged by storage at ambient temperatures from plus 160°F to minus 30° F, after which they shall operate as specified in 3.4.1.

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3.4.3 Fuels, lubricants, and water immersion. Unless otherwise specified (see 6.2), a filter-coalescer element and prefilter element after being immersed in specified fuels and lubricants (see 3.9.1), for 100 hours, and after immersion in saltwater and freshwater for 72 hours, shall show no evidence of swelling, corrosion, Reparation of components, dissolving of adhesives, or deformation which could cause failure during operation.

3.4.4 Post-environmental. Unless otherwise specified (see 6.2), filter-coalscer elements and prefilter elements after being subjected to the environmental conditions specified in 3.4, shall operate without any failure when installed in a separator and subjected to a l-percent by volume No. 2 diesel fuel separation test at rated flow.

3.5 Safety. Unless otherwise specified (see 6.2), all electrical components of the separator automatic control systems will be explosionproof and classified as Class 1, Division 1, Group D, in accordance with Article 500 of the National Fire Protection Association. All electrical systems shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazards.

3.6 Human factors engineering. The separators shall conform to the human factors engineering criteria as specified in MIL-STD-1472.

3.7 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings: All adjustments, repairs, replacements, or maintenance shall be accomplished using common handtools, and special tools if required.

3.7.1 Maintenance ratio. When specified (see 6.2), the separators shall have a maintenance ratio of no greater than 0.03, when tested as specified in 4.4.6.

3.8 Reliability. When specified (see 6.2), the specified Mean-Time-Btween-Failure (MTBF) shall be 140 hours when the separators are operated it rated capacity and tested as specified in 4.4.5.

3.9 Technical requirements.

3.9.1 Fuels, lubricants, and waters. Unless otherwise specified (see 6.2), materials in the separators, coalescer elements, and prefilter elements shall lot be adversely affected by and shall have no adverse effect on waters, from

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potable water to seawater, containing varing amounts of the following fuels and lubricants, alone or in combination:

VV-G-76	MIL-L-9000
W -F-800	MIL-T-5624
MIL-L-2104	MIL-F-16884
MIL-F-24397	

3.9.2 Effluent contamination limits. The required effluent water and separated oil samples shall conform to the following contamination limit requirements:

(a) oil in water. When analyzed in accordance with 4.5.2.1, the effluent water shall not contain a haze, iridescent appearance, visible sheen, visible oil, and not more than 15 parts per million (ppm) of free (dissolved) oil.
(b) Water in separated oil. When tested in accordance with 4.4.4.5, the separated oil shall not contain more than l-percent free (undissolved) water (see 4.5.2.6) .
(c) Solids. when tested in accordance with 4.4.4.6 and 4.4.4.8, the effluent water shall not contain more than 2 milligrams per liter of solids for coalescer elements, nor more than 1 milligram per liter of solids for prefilter elements (see 4.5.2.3).

3.9.3 Performance, The separator including all controls shall conform to the following performance requirements. The flow rate of the separator shall be as specified in the individual detailed specification or end item specification, and the flow rate shall be continuous or intermittent.

3.9.3.1 Differential pressure. Differential pressure across the separator with new coalescer elements and prefilter elements installed and clean freshwater (see 4.4.2.1) shall not exceed 5 pounds per square inch (psi) at any flow rate up to 100-percent of rated flow when measured across the inlet and outlet couplings (see 4.4.4.4). The differential pressure without the coalescer elements and prefilter elements installed shall not exceed 3 psi.

3.9.3.2 Oil removal. The separator shall remove various fuels, lubricants, and mixtures of fuels and lubricants (hereinafter called “oils”) (see 3.9.1) from clean freshwater (see 4,4.2.1), to the limits specified in 3.9.2(a) at waterflow rates up to 100-percent of rated flow, with the influent water containing up to 100-percent by volume of oils. The separated and discharged oil shall contain no more than 1 percent water by volume (see 3.9.2(b)). During the 100-percent oil injection period (see 4,4.4.5), no oil shall pass to the final coalescer stage.

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3.9.3.3 Solids removal coalescer elements only. When tested as specified in 4.4.4.6, the separator (with the prefilter vessel removed), shall remove AC Test Dust (fine) and l-percent by volume of No. 2 diesel fuel to the limits specified in 3.9.2(a) and 3.9.2(b) at waterflow rates up to 100-percent of rated flow.

3.9.3.4 Solids retention capacity, coalescer elements only. When tested as specified in 4.4.4.7, the separator (with the prefilter vessel removed), shall remove and retain a quantity of AC Test Dust (fine) equal to 16 grams (0.6 ounces) per gallon per minute (gpm) of its rated flow capacity. The differential pressure across the first coalescer stage measured at the inlet and outlet couplings at rated flow shall not exceed 15 psi before 30 minutes nor 25 psi before 60 minutes at a solids injection rate of 0.143 grams per gpm and l-percent by volume of No. 2 diesel fuel. At the end of 60 minutes, the diesel fuel injection shall be stopped, and the solids injection rate increased to 0.4 grams per gpm of' rated flow and continued until a 65 psi differential pressure is reached. The 65 psi differential shall be maintained for 5 minutes. No structural failure or deformation of the separator or elements shall be evident.

3.9.3.5 solids removal, prefilter element only. When tested as specified in 4.4,4.8, the separator (with the prefilter element (inside-to-outside or outside-to-inside flow) installed), shall remove AC Test Dust (fine) with up to l-percent by volume No. 2 diesel fuel to the limits specified in 3.9.2(c) at waterflow rates up to 100-percent of rated flow.

3.9.3.6 Solids retention capacity prefilter and prefilter element only. The prefilter (inside-to-outside flow) shall remove and retain a quantity of AC Test Dust (fine) at least equal to 16 grams (0.6 ounces) per gpm of its rated flow capacity. The pressure differential across the prefilter measured it the inlet and outlet couplings at rated flow shall not exceed 15 psi before 30 minutes, nor 25 psi before 60 minutes at a solids injection rate of 0.143 grams per gpm and l-percent by volume of No. 2 diesel fuel. The element with outside-to-inside flow shall retain 16 grams per gpm of its rated flow capacity. Pressure differential shall not exceed 10 psi before 30 minutes, nor 15 psi before 60 minutes at the above injection rates. When tested as specified in 4.4.3.9, at the end of 90 minutes (for either type prefilter element), the diesel fuel injection shall be stopped, and the solids injection rate increased to 0.4 grams per gpm of rated flow and continued until a 65 psi differential is reached. The 65 psi differential shall be maintained for 5 minutes. No structural failure or deformation of the prefilter or elements shall be evident.

3.9.3.7 Life test. When specified (see 6.2), the separator with prefilter installed shall remove AC Test Dust and from 1-percent up to 5-percent by

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volume of No. 2 diesel fuel at rated flow, from clean freshwater, over a period of up to 125 hours to the limits specified in 3.9.2.

3.9.4 Separator vessels. The separator pressure vessels, including prefilter vessels, shall be as specified in the individual detailed specification or in the end item specification. The pressure vessels shall be designed and built in accordance with the requirement of the ASME Boiler and Pressure Vessel Code (Section VIII, Division 1) and need not be stamped. Unless otherwise specified (see 6.2), the pressure vessels shall withstand a hydrostatic Pressure of 1-1/2 times their specified maximum working pressure for a period of 5 minutes without any evidence of structural failure or permanent deformation.

3.9.4.1 Deck plates. Deck plates, if used, in each pressure vessel that form a pressure boundry such that the inlet compartment shall be at least twice the thickness required for structural reasons to provide additional allowance for corrosion. Separate compartments within the vessels such as the inlet compartment when hydrostatically tested to 50-percent of the hydrostatic pressure specified for the vessel for a period of 10 minutes shall show no evidence of structural failure or permanent deformation.

3.9.4.2 Vessel viewports. Unless otherwise specified (see 6.2), each filter-coalescer vessel shall contain two 2-inch diameter tempered-glass viewports located 180-degrees apart, about 2 inches above the lower element mounting surface.

3.9.4.3 Vessel closures. All pressure vessel closures shall be designed for quick opening and closing. If closures are hinged, a holdback device will be provided to prevent accidental closing.

3.9.4.4 Water drain valves. Each pressure vessel shall be equipped with water drain valves at the bottom and block valves between each vessel and the inlet to facilitate draining each vessel independently.

3.9.4.5 Automatic air vents. The prefilter vessel shall be equipped with an automatic air-venting device in the quick-opening closure. When specified (see 6,2), coalescer vessels shall also be equipped with automatic air-venting devices. Each vent valve should be provided with a length of hose sufficient long to reach ground or deck level.

3.9.4.6 oil-water interface sight gage. Each pressure vessel shall be equipped with an external, unbreakable glass sight gage that will show the oil-water interface level in the upper half of the vessel.

3.9.4.7 Internal oil discharge piping. The internal oil discharge piping shall be configured to permit discharge of essentially all of the oil

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accumulated in the top of each pressure vessel without interfering with removal of the quick-opening closures or replacement of the expendable elements and prefilters.

3.9.4.8 Sampling valves. In-line sampling valves shall be provided in the inlet piping to each vessel, in the oil discharge line, and downstream from the water discharge valve.

3.9,4,9 Pressure indicators. Pressure indicators conforming to MIL-I-18997 shall be provided to indicate system inlet pressure and the inlet and outlet pressure of each vessel. Indicators shall not have movable parts exposed to seawater.

3.9.4.10 Automatic oil discharge system. Unless otherwise specified (see 6.2), the prefilter vessel and the first coalescer vessel shall be equipped with automatic oil discharge systems. When a separator system is so equipped, it shall also be equipped with an automatic water discharge control valve downstream from the final coalescer vessel. The system shall be capable of operating with the influent water containing up to 100-percent oil (see 4.4.4.5).

3.9.4.10.1 System controls. When equipped with automatic controls, each vessel shall be equipped with a capacitance-type probe that controls activation of the automatic oil discharge cycle. Unless otherwise specified (see 6.2), other type controls maybe furnished provided they meet the requirements of 4.4.3.5. In all cases, water and oil valves shall automatically fail in the closed position when the power source fails or is shut off.

3.9.4.10.2 Power requirements. The automatic oil discharge system specified in 3.9.4.10.1 requires approximately 15 pounds per square inch gage (psig), air supply. The systems shall be available for use with 115 V ac, 60 Hertz 120 V dc, and 24 V dc electrical power supplies as specified (see 6.2), The electromagnetic interference characteristics requirements for this system shall conform’ to MIL-STD-461, Class II B, and the system shall be installed in accordance with IEEE Standard No. 45 for shipboard applications when applicable.

3.9.4.11 Physical requirements. The overall dimensions and weight of the separator shall be as specified in the individual detailed specification or the end item specification. All pressure vessels shall have a minimum inside diameter of 10 inches. For shipboard application, each component of the system shall be capable of being passed through an opening 26 inches by 48 inches without cutting or rewelding of the component.

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3.9.5 Filter-coalescer elements. The filter-coalescer elements shall be the standard expendable military element conforming to MIL-F-52847. Each element shall remove solids and coalesce oils into large droplets for flow rates specified for their dimensions. The elements shall withstand a differential pressure of 65 pounds per square inch differential (psid), without structural failure or permanent deformation.

3.9.6 prefilter elements. Prefilter elements shall conform to 3.9.6.1 or 3.9.6.2 as specified (see 6.2).

3.9.6.1 Prefilter elements, outside-to-inside flow. Prefilter elements, outside-to-inside flow, shall be a standard expendable military prefilter element of the pleated-paper type, 10-micron nominal rating, conforming to MIL-F-52847.

3.9.6.2 Prefilter elements, inside-to-outside flow. Prefilter elements, inside-to-outside flow, shall be a standard expendable military prefilter element of the pleated-paper/fiberglass type, 5 micron nominal rating, conforming to MIL-F-52847.

3.10 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.10.1), with l-percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.10.1 Composition. Composition I shall be used unless Composition II must be used in order to comply with local air pollution regulations in the application of the varnish. When Composition II in used, the supplier shall provide evidence to the Government that the use of Composition IX is required, and shall certify that the Composition II material complies with Rule 66 - Air Pollution Control District, County of Los Angeles, CA.

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3.11 Slinging provisions. When required by the individual detailed specification or end item specification, permanently affixed slinging provisions that enable the separator to be lifted in its normal travel position shall be provided and shall conform to MIL-sTD-209.

3.12 Identification markings. The separators shall be identified in accordance with MIL-STD-130. The marking shall be applied to the separator on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached and located in accordance with the individual detailed specification or end item specification.

3.13 Instruction plates. Each separator shall be equipped with instruction plates or diagrams, including warnings or cautions, describing any important or special procedure to be followed in assembling, operating, or servicing the separator. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached and located in accordance with the individual detailed specification or end item specification.

3.14 Treatment and painting. The portions of the separator normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type B. The color shall be deck gray In accordance with JAN-P-700 for shipboard applications, or color as specified (see 6.2), for land-based applications.

3.15 Dissimilar metals. Unless otherwise specified, the selection of materials and corrosion prevention due to galvanic couples due to dissimilar metals shall be in accordance with MIL-T-704 and MIL-STD-889.

3.16 Government-furnished property. The quantity and type of filter-coalescer and prefilter elements conforming to MIL-F-52847 required by the individual detailed specification or end item specification will be furnished by the Government (see 6.6).

3.17 Workmanship.

3.17.1 Metal fabrication. Metal used in fabrication shall be free of kinks and sharp bends. The straightening of materials shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.17.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter other than mill scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be required to enable the separators to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the separators shall be at the option of the supplier.

3.17.3 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code, Section 5, Qualification.

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Suppliers who only make horizontal welds need not qualify welders for “all position welding”. Subject to approval by the Government, the supplier’s standard welder qualification may be substituted in lieu of the above codes provided that the supplier’s procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified In the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The supplier shall maintain sufficient inspection records on each system produced to permit the Government to determine compliance with inspection requirements.

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4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as specified in the individual detailed specification or end item specification.

4.3 Qualification or preproduction inspection. Examination and tests shall be in accordance with the individual detailed specification or end item specification.

4.4 Quality conformance inspection.

4.4.1 Sampling, examination and tests. Sampling for examination and tests shall be as specified In the individual detailed specification or end item specification.

4.4.2 Test conditions.

4.4.2.1 Test water. Fresh, prefiltered tapwater shall be used in all tests. The freshwater shall contain no more than 1.0 milligram per liter of solids (see 4.5.2.3), and shall have a surface tension (see 4.5.2.4), of not less than 65 dynes/centimeter at 75° I?. The acidity-alkalinity value (pH) (see 4.5.2.5), shall be between 5 and 8. Test water shall be used on a once-through, single-pass basis and not recirculated during testing.

4.4.2.2 Water temperature. Unless otherwise specified (see 6.2), the test water temperature shall be between 50° F and 80° F during the testing specified herein, and shall be mainitained within plus or minus 5° F from the starting temperature.

4.4.2.3 Test contaminants. The following contaminants shall be used:

(a) Siliceous dust (AC Test Dust), obtainable from AC Spark Plug Division, 1300 North Dort Highway, Flint, Michigan 48504, identified as AC Spark Plug Company test duet, fine grade, package No. 1543094. (Particle size (microns) distribution, percent by weight? 0-5, 39; 0-10, 57; 10-20, 16; 20-80, 27).

(b) Contaminant oils. Unless otherwise specified (see 6.2), No. 2 diesel fuel conforming to VV-F-800, or marine diesel fuel conforming to MIL-F-16884 shall be used at the discretion of the supplier or test activity. Lubricating oil conforming to MIL-L-2104, Grade 30 or MIL-L-9000 (Symbol 9250) shall be used at the discretion of the supplier or test activity. All contaminant oils shall be used on a once-through, single-pass basis and not reused for testing.

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4.4.2.4 Test data. Test data shall be recorded as shown in Figure 2 for each test as stated in the applicable individual tests.

4.4.2.5 Waterflow rate. All full scale tests shall be performed at 100 percent of the flow rating of the separator, coalescer elements, or prefilter elements.

4.4.3 Separator test loop system. Arrangement of the separator test loop system shall be as shown In Figure 1 and as specified herein.

4.4.3.1 Instruments. Instruments such as meters, gages, and thermometers used in the test loop shall be laboratory precision type and shall be calibrated at intervals spaced to assure laboratory accuracy.

4.4.3.2 Main supply pump. The main supply pump shall be as specified in the individual detailed specification or end item specification depending on specified flow rate and end item application. Unless otherwise specified (see 6.2), the supply pump used for single-element coalescer and prefilter element testing shall be a double-diaphragm air-operated pump of appropriate capacity equipped with a suitable pulsation dampener, except that a centrifugal pump operating at 3450 revolutions per minute (rpm), shall be used for the third hour test specified in 4.4.4.5.

4.4.3.3 Waterflow meter. A meter shall be provided to indicate the specified waterflow rate within an accuracy range of 1.0 percent at rated flow.

4.4.3.4 Fuels and lubricants flow meters. Meters shall be provided to indicate the injection rate of the fuels and lubricants added to the water as contaminants. The meter or meters shall be calibrated for the specific gravity of the specified fuels and lubricants, and the accuracy range shall be within plus or minus 2-percent of the specified injection flow rate. A calibrated supply tank may be used for viscous oils.

4.4.3.5 Solid contaminant feeder. A continous solid contaminant feeder with an accuracy range within plus or minus l-percent of the specified feed rate shall be provided to feed the solid contaminants into the test loop system.

4.4.3.6 Test separator. Unless otherwise specified (see 6.2), a single element, 5-gpm, 10-gpm or 20-gpm, (as required), 3-stage separator, (2-coalescers, 1-prefilter), that has successfully passed all tests specified herein, shall be provided for the testing of filter-coalescer elements and prefilter elements.

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4.4.3.7 Cleanup separator. The cleanup separator shall be of a type that has successfully passed all tests specified herein, and shall be capable of processing water at the highest rate of flow required in the test loop system.

4.4.3.8 Differential and other pressure gages. Pressure gages shall be provided for installation in the test loop system. One gage shall be installed between the main supply pump and the prefilter, and another shall be installed downstream from the final coalescer stage at the water discharge. A third pressure gage may be installed in the oil discharge line just ahead of the slop-oil receiving tank. Differential pressure gages connected to the inlet and outlet couplings of each of the 3 stages may be used, or pressure gages located in the piping between the prefilter and first coalescer, and between successive coalescer stages will be acceptable. The accuracy of any gage used shall be such that any error shall not exceed plus or minus 2 psig.

4.4.3.9 Turbidity meter. The test loop shall be provided with a full-flow water quality monitor, precalibrated in ppm, installed on the water effluent side of the test separator. The monitor shall be a Keene Corporation, (formerly Bowser-Briggs), turbidimeter, Model 861 B, Mark II, or equal (see 6.5). A Fisher-Porter Company, Model 2202 BL electric chart recorder, or a Rustrak Instrument Company, Inc., miniaturized automatic chart recorder, Model 88, may be used at the discretion of the testing activity.

4.4.3.10 Water filter. A water filter shall be provided for filtering the test water to the cleanliness specified In 4.4.2.1.

4.4.3.11 Sampling devices. Probe-type sampling devices, facing upstream shall be installed in the test loop as shown in Figure 1, within 10 pipe, (or hose), diameters of the separator outlet or inlet as applicable.

4.4.3.12 Test loop system piping. The flow system piping between the supply pump and the test separator shall be the same diameter as the piping on the test separator. The distance between the pump and separator shall be kept to a minimum. Solids shall be injected as close as practicable to the test separator, and shall not be passed through the supply pump. Oil contaminants shall be injected into the suction side of the supply pump. Hardwall hose of equivalent size is acceptable in lieu of metal pipe.

4.4.4 Tests. Unless otherwise specified in the individual detailed specification or end item specification, all tests shall be as specified herein.

4.4.4.1 Filter-coalescer element and prefilter element, inside-to-outside Flow. The filter-coalescer elements and prefilter elements (inside-to-outside Flow, shall be subjected to the following test:

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(a) Equipment. A dummy element shall be used to isolate the inlet compartment. Torque shall not exceed 27 foot-pounds when installing the dummy. An open tank of prefiltered freshwater, a pump, flowmeter, pressure gage, piping or flexible hose, valves, fixtures for retaining the elements, and metered supply of No. 2 diesel fuel, and a fuel drain are required for this test. The retaining fixture shall support the elements in the same manner as the vessel for which the elements are intended, and shall direct the flow in the normal direction through the element. A schematic arrangement Of the element test equipment is shown in Figure 3.

(b) Procedure. The test shall be conducted as follows:

(1) Submerge mounted element in the water.
(2) Start waterflow and gradually increase flow rate, expelling all air, until rated flow for the element is reached.
(3) Inject No. 2 diesel fuel at lo-percent of rated flow until fuel droplets are formed on the surface of the element.
(4) Rotate the element, and observe the fuel droplets and water. The formation of clusters of fuel droplets, graping, or cloudy streams shall constitute failure of this test.

4.4.4.2 Prefilter element, outside-to-inside flow. The prefilter elements (outside-to-inside flow), shall be sub jetted to the following test:

(a) The equipment required shall be as specified in 4.4. 4.1(a) and as shown in Figure 3.
(b) Procedure. The test shall be conducted as follows:

(1) Support the fixture and element with the outlet just below the surface of the water.
(2) Start the water flow and gradually increase the flow rate, expelling all air, until rated flow is reached.
(3) Inject No. 2 diesel fuel at 10-percent of rated flow until fuel droplets are visible in the water. Note clarity of water and fuel. To determine the clarity, a small sample of the water discharged from.the element-holding case shall be collected in a clean transparent bottle or beaker and observed for condition of the fluid. Evidence of unclear water or no definitive interface between the water and fuel shall constitute failure of’ this test.

4.4.4.3 Hydrostatic pressure test, pressure vessels. The separator pressure vessels shall be subjected to a hydrostatic test pressure as specified In the individual detailed specification or end item specification. When not

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specified in the individual detailed or end item specification, the hydrostatic pressure tests shall determine conformance to the requirements specified in 3.9.4 of this specification. Nonconformance to the requirements in the Individual detailed or end item specification or 3.9.4 shall constitute failure of this test.

4.4.4.3.1 Hydrostatic pressure test, Inlet compartment. The separator pressure vessels containing deck-plates and inlet compartment shall be subjected to a hydrostatic test pressure as specified in the individual detailed or end item specification. When not specified in the individual detailed or end item specification, the hydrostatic pressure test shall determine conformance to the requirements specified in 3.9.4.1 of' this specification. Nonconformance to the requirements in the individual detailed or end item specification or 3.9.4.1 shall constitute failure of this test.

4.4.4.4 Differential pressure (first set of elements and prefilters). Using a new set of dry elements and prefilters which have not been subjected to any other tests, the separator shall be subjected to a flow test at the rates shown in Table I, to determine the pressure drop across the pressure vessels, coalescer elements, and prefilter elements. Pressure drops across all stages in the separator shall be recorded at the end of each period. Test data shall be recorded as shown in Figure 2. Nonconformance to 3.9.3.1 shall constitute failure of this test.

Table I. Rate of Flow for 60 Minutes

Period	Time (Minutes)	Percent of Rated Flow
1	0 to 10	100
2	10 to 20	80
3	20 to 30	60
4	30 to 40	40
5	40 to 50	20
6	50 to 60	100

4.4.4.5 Oil removal. The oil removal tests (see 3.9.3.2), shall consist of three separate 1-hour runs. Unless otherwise specified in an individual detailed specification or end item specification, oils listed below shall be introduced at the following concentrations:

First Hour: 1-percent by volume lubricating oil.
Second Hour: 5-percent by volume of mixture of 5 parts diesel fuel and 1 part lube oil.
Third Hour: 25-percent by volume diesel fuel.

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Unless otherwise specified (see 3.9.4.10), pump suction will be stitched from the freshwater supply to the slug-test reservoir during the last 15 minutes of the third-hour tests, The reservoir shall be partially filled with freshwater with a layer of the fuel-lube oil mixture used in the second-hour test on top. The relative amount of each will depend on the flow rate of the test separator. Taking suction at the bottom of the reservoir, pumping shall continue from 100-percent water through the interface to 100-percent oil. After sufficient oil has been pumped through the test separator to prove operability of the automatic oil discharge system, water shall be introduced into the reservoir to permit pumping 100-percent water. After allowing 3 minutes for the test separator to recover after the 100-percent oil slug, the effluent water should meet the cleanliness requirements of 3.9.2(a). After 30 minutes of each 1-hour test, a 1-liter sample of the discharged oils shall be collected and tested for water. A separate test data sheet (see Figure 2), shall be used for each l-hour test. Nonconformance to 3.9.2(a), 3.9.2(b), or 3.9.3.2 shall constitute failure of this test.

4.4.4.6 S0lidS removal, coalescer elements only. The input water shall be continuously contaminated with AC Test Dust at the rate of 0.143 grams per gpm of waterflow rate and l-percent by volume of No. 2 diesel fuel for a period of 15 minutes (see 3.9.3.3). A test data sheet (see Figure 2), shall be used to record data. Samples shall be taken for solids analysis at the end of the first 5 minutes and during the last 5 minutes (see 4.5.2.3). Nonconformance to 3.9.2 shall constitute failure of this test.

4.4.4.7 Solids retention capacity, coalescer elements only (same elements as 4.4.4.6). During the solids retention capacity test (see 3.9.3.4) , the input water shall be continuously contaminated with 0.143 grams of AC Test Dust per gpm of water and 1-percent by volume of No. 2 diesel fuel. The test shall continue until a 25 psi differential pressure is reached across the separator. The differential pressure across the first coalescer stage should not exceed 15 psi before 30 minutes nor 25 psi before 60 minutes. When installed, the differential across the second coalescer stage should not exceed 5 psi before 60 minutes. After 60 minutes, the fuel injection shall be stopped and the solids injection rate increased to 0.4 grams per gpm of rated flow and continued until a 65 psi differential is reached. The 65 psi differential shall be maintained for 5 minutes. A test data sheet (see Figure 2), shall be used to record the data. samples shall be taken for solids analysis at the end of 30, 60, and 80 minutes (see 4.5.2.3). Nonconformance to 3.9.2 or 3.9.3.4 shall constitute failure of this test.

4.4.4.8 solids removal prefilter elements only. The input water shall be continuously contaminated with AC Test Dust at the rate of 0.143 grams per gpm

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of waterflow rate and 1-percent by volume of No. 2 diesel fuel for a period of 15 minutes (see 3.9.3.5). A test data sheet (see Figure 2), shall be used to record data. Samples shall be taken for solids analysis at the end of the first 5 minutes and during the last 5 minutes (see 4.5.2.3). Nonconformance to 3.9.2(c) shall constitute failure of this test.

4.4.4.9 Solids retention capacity, prefilter and prefilter elements only (same elements as 4.4.3.8). During the solids retention capacity test (see 3.9.3.6), the input water shall be continuously contaminated with 0.143 grams of AC Test Dust per gpm of water and l-percent by volume of No. 2 diesel fuel. For the inside-to-outside flow element, the test shall continue until a 25 psi differential pressure is reached across the prefilter. The differential pressure should not exceed 15 psi before 30 minutes nor 25 psi before 60 minutes. For the outside-to-inside flow element, the test shall continue until a 15 psi differential pressure is reached across the prefilter. The differential pressure should not exceed 10 psi before 30 minutes nor 15 psi before 60 minutes. At the end of 60 minutes (for either type element), the fuel injection shall be stopped, and the solids injection rate increased to 0.4 grams per gpm of rated flow and continued until a 65 psi differential is reached. The 65 psi differential shall be maintained for 5 minutes. A test data sheet (see Figure 2), shall be used to record the data. Samples shall be taken for solids analysis at the end of 30, 60, and 80 minutes (see 4.5.2.3). Nonconformance to 3.9.2(c) or 3.9.3.6 shall constitute failure of this test.

4.4.4.10 Life test (second set of elements). Water shall be pumped through the separator at rated input flow for up to 125 hours. Test data sheets (see Figure 2), shall be used to record the data. Nonconformance to 3.9.3.7 shall constitute failure of the test. Test conditions shall be as follows:

(a) The life test may be accomplished in a maximum of 16 operating increments of 8 hours per day for 15 consecutive work days and a final operating increment of 5 hours on the 16th day. Start-up and shutdown shall be accomplished by starting and stopping of the supply pump.
(b) No. 2 diesel fuel shall be injected at the beginning of each operating increment for 1 hour or during the time solids are added, whichever is longer. The rate of fuel injection shall be 1-percent by volume, except for the final 1 hour (125th hour), during which time the rate of fuel inspection shall be increased to 5-percent by volume.
(c) Solid contaminates shall be AC Test Dust (fine), added dry at the rate of 0.143 grams per gpm of flow rate at the beginning of each operating increment as required to produce and maintain a pressure differential of 10 psi across the separator. The start of solids addition shall be immediately after the start of fuel inJection.

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4.4,4.11 Environmental. When specified (see 6.2), the environmental tests (see 3.4), shall be conducted in the order listed in the following paragraphs, on a complete separator, coalescer elements, and prefilter elements, In addition, samples of gasket material, O-ring seals, and other system components that may be subject to environmental damage shall also be subjected to these tests.

4.4.4.11.1 Operation, high temperature (vessels). When specified (see 4.4.4.11), install the complete separator including correct number of coalescer and prefilter elements in a pumping system of the required flow rate. Operate the separator at rated flow in an ambient temperature of plus 125° F for a period of not less than 1 hour. Nonconformance to the high temperature operating requirements specified in 3.4.1 shall constitute failure of this test.

4.4.4.11.2 Operation, low temperature (vessels). When specified (see 4.4.4.11), install the complete separator as specified in 4.4.4.11.1. Operate the separator at rated flow in an ambient temperature of plus 33° F for a period of not less than 1 hour. Nonconformance to the low temperature operating requirements specified in 3.4.1 shall constitute failure of this test.

4.4.4.12 Storage, high temperature (SHT). When specified (see 4.4.4.11), one of the following two tests shall be made applicable.

4.4.4.12.1 SHT, separator vessels. Install new coalescer and prefilter elements and subject the complete separator to test in accordance with MIL-STD-810, Method 501, Procedure 1, Steps 1, 2, 3, 4, and 8. Nonconformance to the high temperature storage requirements specified in 3.4.2 shall constitute failure of this test.

4.4.4.12.2 SHT, coalescer and prefilter elements (fifth set of elements). Test the new coalescer and prefilter elements in accordance with MIL-STD-810, Method 501, Procedure 1, Steps 1, 2, 3, and 8. Nonconformance to the high temperature storage requirements specified in 3.4.2 shall constitute failure of this test.

4.4.4.13 storage, low temperature (SLT). When specified (see 4.4.4.11), one of the following two tests shall be conducted as applicable.

4.4.4.13.1 SLT, separator vessels. Install new coalescer and prefilter elements and test the complete separator in accordance with MIL-STD-810 Method 502, Procedure 1, Steps 1, 2, 3, and 4. Storage temperature shall be minus 30° F for not less than 48 hours. Nonconformance to the low temperature storage requirements specified in 3.4.2 shall constitute failure of this test.

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4.4.4.13.2 SLT, coalescer and prefilter elements. Subject the same coalescer and prefilter elements, which were tested in 4.1.4.12.2 to the low temperature test in accordance with MIL-STD-810, Method 502 Procedure 1) Steps 1, 2, and 3. Storage temperature shall be minus 30° F for not less than 48 hours. Nonconformance to the low temperature storage requirements specified in 3.4.2 shall constitute failure of this test.

4.4.4.14 Resistance to fuels coalescer and prefilter elements). When specified (see 6.2), the following fuels resistance tests shall be conducted using one or more of the fuels and lubricants in 3.9.1 as specified in the individual detailed or end Item specification. Using the elements from tests in 4.4.4.12.2 and 4.4.4.13.2, establish the fuel resistance of these elements by immersing in the specified test fuels for a minimum period of 100 hours at a temperature of 77° F, plus or minus 5° F, and examine for defects. Nonconformance to the fuel immersion requirements specified in 3.4.3 shall constitute failure of this test.

4.4.4.15 Resistance to water (coalescer and prefilter elements}. When specified (see 6.2), establish the water resistance (fresh and salt) of the elements from tests specified in 4.4.4.14 by immersion in freshwater (see 4.4.2.1) and in a solution consisting of 4 percent sodium chloride and 96 percent distilled water for a period of 72 hours, and examine for defects. Nonconformance to water immersion requirements specified in 3.4.3 shall constitute failure of this test.

4.4.4.16 Post-environmental performance (coalescer and prefilter elements). After successful completion of the tests specified in 4.4.4.12.2, 4.4.4.13.2, 4.4.4.14, and 4.4.4.15, install the same elements in a test separator and subject these elements to a l-percent by volume No. 2 diesel fuel test. Nonconformance to 3,4.4 shall constitute failure of this test.

4.4.5 Reliability. When specified (see 3.8), the initial production separator(s) shall be tested as specified in 4.4.4.5 and 4.4.4.10. This test cycle shall be repeated until 550 hours of operation have been accumulated, For testing purposes, a failure is defined as any malfunction which the operator cannot correct within 30 minutes by adjustment, repair, or replacement action using controls, on-equipment tools or parts, and which causes or may cause:

(a) Failure to commence operation, cessation of operation or degradation of performance below specified levels.
(b) Damage to the separator by continued operation.
(c) A serious personnel safety hazard.

Occurrence of six or more failures shall constitute failure of this test.

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4.4.6 Maintenance ratio. When specified (see 3.7.1), the maintenance ratio shall be computed during testing as specified in 4.4.5. The maintenance ratio is defined as the total active maintenance man-hours (scheduled and unscheduled) required to the total operating time. Daily operational checks are excluded. Nonconformance to 3.7.1 shall constitute failure of this test,

4.5 Sampling containers.

4.5.1 Preparation of sampling containers. The sample containers used for analytical determinations shall be glass with screw-type caps. The caps shall be lined with a metal foil. The following procedure has been found suitable for container preparation:

(a) Wash the container and cap thoroughly with hot water and detergent and rinse thoroughly in hot tapwater.
(b) Rinse twice with particle-free distilled water.
Rinse with particle-free isopropyl alcohol to remove water.
(d) Hold bottle in an inverted position while flushing with filtered air from a Milipore aerosol holder until the alcohol has evaprated, and cap bottle.

4.5.2 Analysis of samples.

4.5.2.1 Free oil detection. The analysis of effluent water for free oil shall be made using the Keene Corporation Model 861-B, Mark 11 Turbidimeter or equal. The pickup head shall be installed in the vertical position with the direction of flow from bottom to top. The turbididimeter shall be the determining factor as to conformance or nonconformance of free oil limitation. Calibration instructions for the turbidimeter shall be in accordance with the manufacturer's standard practice.

4.5.2.2 Free oil detection pyrographic analysis. As a spotcheck for the presence of free oil in the effluent water, the pyrographic analyzer, Mark III, located at the U.S. Army Mobility Equipment Research and Development Center, ATTN: STSFB-GFF, Fort Belvoir, Virginia may be used. Samples of effluent water taken for analysis should be properly prepared (see 4.5.1), and labeled with the type of fuel or lubricant and water used in the particular test.

4.5.2.3 Solid contaminants. The analysis of the l-liter samples shall be In accordance with Appendix I of this specification. Nonconformance to 3.9.2(c) shall constitute failure of this test.

4.5.2.4 Surface tension. Surface tension of the test water shall be performed as specified in ASTM D1331, Method A. Nonconformance to 4.4.2.1 shall constitute failure of this test.

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4.5.2.5 pH determination. The pH of the test water shall be determined by using a direct reading instrument such as a Leed and Northrup Model 7401 pH meter or approved equal. Nonconformance to 4.4.2.1 shall constitute failure of this test.

4.5.2.6 Water in discharged oil. The l-liter sample drawn after the first 30 minutes of each oil removal run shall be allowed to settle for a period of 24 hours prior to determining the volume of water. Nonconformance to 3.9.2(b) shall constitute failure of this test.

50 PREPARATION FOR DELIVERY

(See Individual end item or detailed specification).

6. NOTES

6.1 Intended use. The separators covered in this specification are intended for use in conjunction with individual detailed specification or end item specification for procurement and testing of oil-water separators, filter-coalescer elements, prefilter elements, and other components of oil-water separation systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification; individual detailed or end item specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2),
(d) Operating and storage temperatures if other than as specified (see 3.4.1 and 3.4.2).
(e) Types of fuels, lubricants, and water to be used for element immersion tests if other than as specified (see 3.4.3).
(f) Post-environmental tests required if other than as specified (see 3.4.4).
(g) Safety requirements if other than as specified (see 3.5).
(h) When maintenance ratio is required (see 3.7.1).
(i) When Mean-Time-Between-Failure for reliability is required (see 3.8).
(j) When materials are to be compatible with fuels, lubricants, and waters other than those listed in 3.9.1.

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(k) When life test is required (see 3.9.3.7).
(l) Hydrostatic test pressure if other than 1-1/2 times working pressure (see 3.9.4).
(m) Vessel viewports required if other than as specified (see 3.9.4.2).
(n) When additional automatic air-venting devices are required on coalescer vessels (see 3.9,4.5),
(o) Automatic oil discharge system required if other than as specified (see 3.9.4.10).
(p) When other types of controls maybe provided (see 3.9.4.10.1).
(q) Type of power required for automatic system (see 3.9.4.10.2).
(r) Type of prefilter elements required (see 3,9.6).
(s) Color required for land-based applications (see 3.14).
(t) Government-furnished property (see 3.17).
(u) Welding process required if other than as specified (see 3.17.2).
(v) Water temperature for testing if other than as specified (see 4.4.2.2).
(w) Contaminant oils to be used if other than as specified (see 4.4.2.3(b)).
(x) Type of supply pump and test separator to be wed for testing if other than as specified (see 4.4.3.5 and 4.4.3.6).
(y) When environmental testing is required (see 4.4.4.11).
(z) When fuel resistance tests are required (see 4,4.4.14).
(aa) When water resistance tests are required (see 4.4.4.15).

6.3 Qualification. The separators covered by this specification include qualified components (coalescer and prefilter elements) or materials, and the contracting officer should include the requirements of ASPR 1-1107.2(b) in the solicitation. When coalescer elements and prefilter elements are submitted to a Government laboratory for qualification, the responsible activity will require the supplier to furnish at the same time a complete set of test data (see Figure 2), covering the test results obtained from tests conducted at his Own facility.

6.4 Preproduction model. Any changes or deviations of production separators from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish separators conforming to this specification.

6.5 Turbidimeter. When a turbidity meter is furnished on an “or equal” basis, it shall be subject to approval by the contracting officer (see 4.4.3.9).

6.6 Government-furnished property, The contracting officer should arrange to furnish the property specified in 3.16.

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6.7 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each separator.

6.8 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of separators.

6.9 Quality conformance. When a supplier is required to submit sample coalescer or prefilter elements to the Government for quality conformance inspection, such sampling should be specified in MIL-F-52847 or the individual detailed or end item specification as applicable.

Custodian:

Army - ME
Navy - SH
Air Force - 84

Preparing activity:

Army - ME

Project No. 4330-0125

User activities:

Army - TM
Navy - CG

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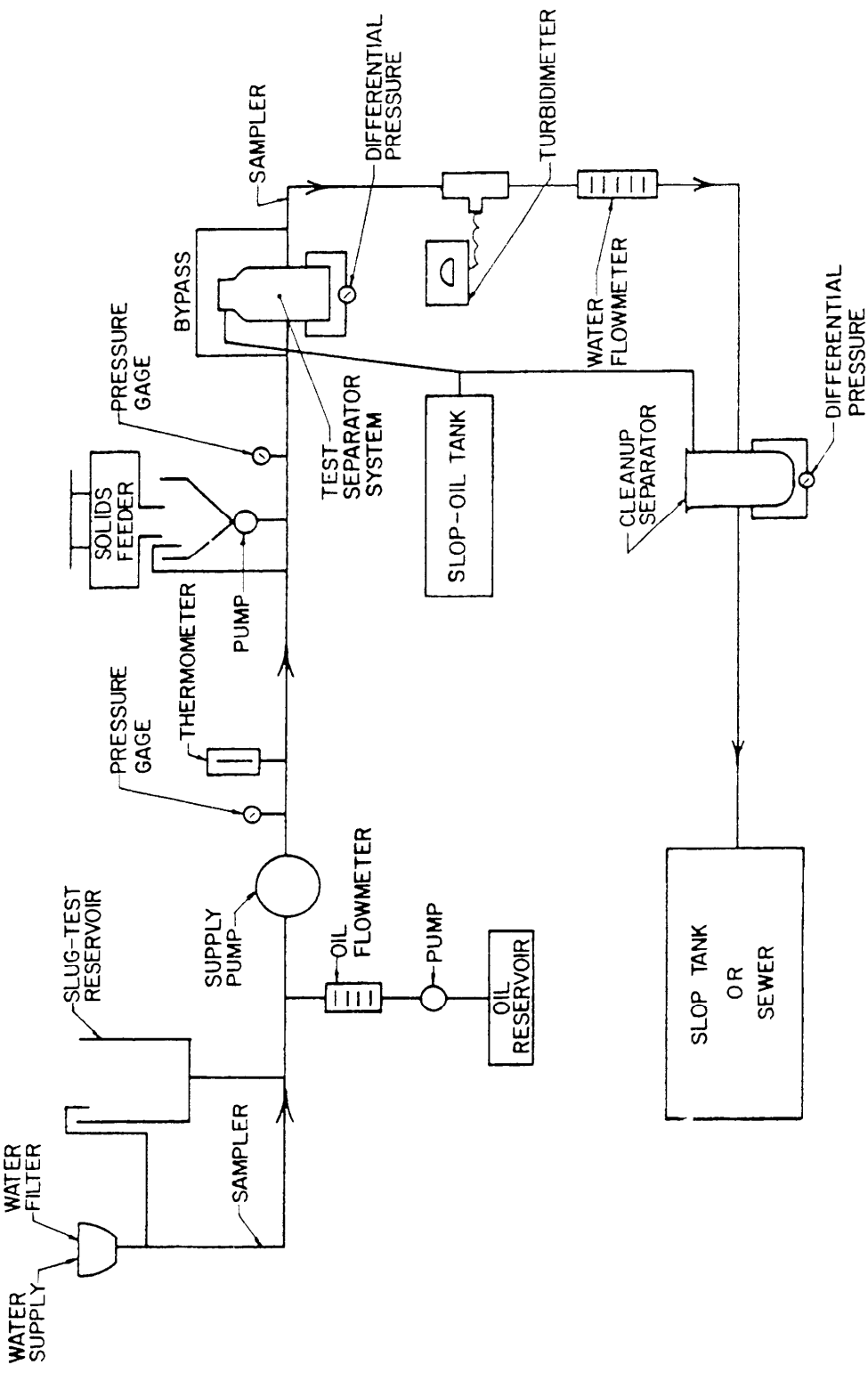


FIGURE 1. ARRANGEMENT OF TEST LOOP SYSTEM

X-2951

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Mfg; _____ Contract Number: _____

Set of Elements: _____ Date of Test: _____ Government Inspector: _____

Type of Test: _____ Type Elements: _____

Test Personnel: _____ Fuel: Add Rate: _____

Solids: Type: _____ Add Rate: _____ Total Amount Added: _____

Time of Day	Time From Start of test (Min)	Flow Rate (GPM)	Pre-filter		△ P Across PF (PSI)	First Stage		△ P Across FS (PSI)	Second Stage		△ P Across SS (PSI)	H_2O Temp (°F)	Effluent fuel Totamitor Reading		
			Inlet Press (PSI)	Outlet Press (PSI)		Inlet Press (PSI)	Outlet Press (PSI)		Inlet Press (PSI)	Outlet Press (PSI)			Reading No	Results PPM	

Figure 2 - Data Sheet

X-2952

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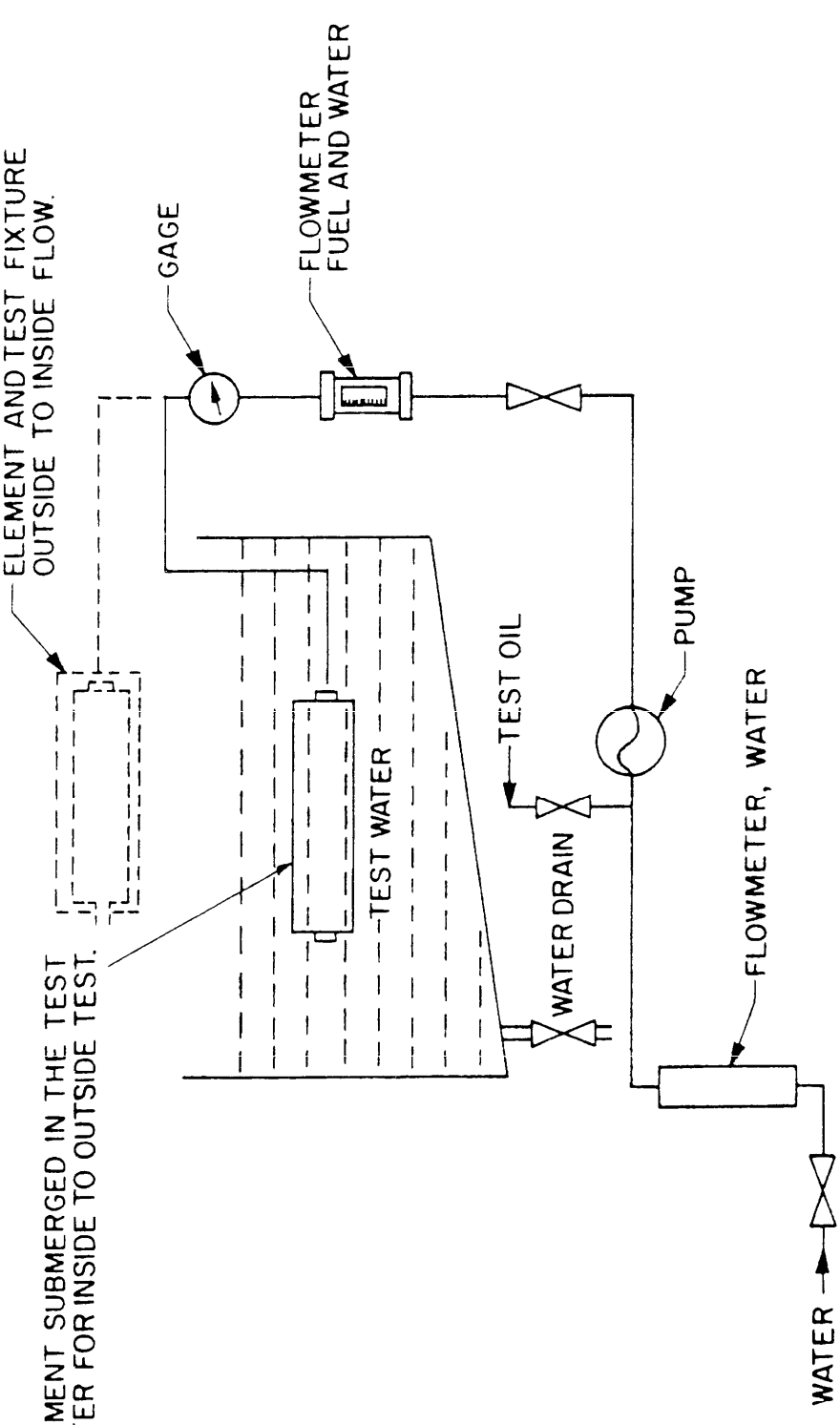


FIGURE 3. FILTER-COALESCER ELEMENT TEST EQUIPMENT. X-2953

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APPENDIX I

FOR

SEPARATORS, OIL-WATER, COALESCER TYPE: FILTER ELEMENTS, FLUID PRESSURE, COALESCER, OILY WASTEWATER, EXPENDABLE: AND PREFILTER ELEMENTS, OILY WASTEWATER, EXPENDABLE

SCOPE. This appendix contains the in-line sampling and analysis procedures for solid contaminants to determine conformance to the requirements of this specification.

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APPENDIX I

In-line Sampling and Analysis Procedures for Solids Contaminant

Apparatus

1. Apparatus for in-line sampling for solids:

   (a) Millipore Sampling Kit, Cat. No. XX64 037 00 (or equal) with legal (U.S.) graduated polyethylene bottle, Cat. No. XX64 037 12; and Matched Weight Monitors, Cat. No. MAWP 037 PM.

2. Apparatus for determination of total contaminant in accordance with ASTM D2276:

   (a) Glassware as required to assemble apparatus as in ASTM Figure.
   (b) Solvent filtering dispenser.

3. Analytical balance.
4. Oven for obtaining 105° C.

Procedure

1. Withdraw sample in accordance with operating Instructions.
2. Flush monitor according to ASTM D2276, using the method which employs flushing the monitor with a solvent filtering dispenser.
3. Remove the filters from the monitor. Note that each monitor contains contains two filters (one test filter on top of one control filter). Place each set of filters into glass petri dishes, using stainless steel forceps. Label dishes.
4. Dry in oven at 105° C for 45 minutes.
5. Remove filters from oven and allow to cool to equilibrate to ambient room conditions for at least 15 minutes. Separate test filter from control filter.
6. Weigh filters.
7. To determine the weight of solids, subtract weight of the control filter from the weight of the test filter. The difference is the weight of total solids.
8. Based upon the sample volume, report the final results in mg/liter.

Note: Available apparatus source - Millipore Filter Corporation, Bedford, Massachusetts.

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DOCUMENT IDENTIFIER AND TITLE MIL-S-52846 Separators, Oil-Water, Coalescer Type: Filter Elements, Fluid Pressure, Coalescer, Oily Wastewater, Expendable: etc.

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